



PROJECT REPORT

**COURSE: LAB211 (Object-Oriented Programming
Lab)**

MOUNTAIN HIKING PROJECT

**Mountain Hiking Registration Management System
Demo**

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1. Main Menu

1.1. Main Menu display

```

-----
| Main menu                                     |
-----
| 0.Exit the Program                           |
| 1.New Registration                           |
| 2.Update Registration Information              |
| 3.Display Registered List                     |
| 4.Delete Registration Information              |
| 5.Search Participants by Name                 |
| 6.Filter Data by Campus                       |
| 7.Statistics of Registration Numbers by Location |
| 8.View List of mountain info                  |
| 9.Save Data to File                           |
|                                              |
-----
Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: |

```

Figure 1: Main menu display

1.2. Input choice

1.2.1 Catch Exception Case / Input Validation

```

-----
| Main menu                                     |
-----
| 0.Exit the Program                           |
| 1.New Registration                           |
| 2.Update Registration Information              |
| 3.Display Registered List                     |
| 4.Delete Registration Information              |
| 5.Search Participants by Name                 |
| 6.Filter Data by Campus                       |
| 7.Statistics of Registration Numbers by Location |
| 8.View List of mountain info                  |
| 9.Save Data to File                           |
|                                              |
-----
Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: -3
Please enter again!
Input your choice: a
Please enter again!
Input your choice: s
Please enter again!
Input your choice: f
Please enter again!
Input your choice: asddd0---09
Please enter again!
Input your choice:

```

Figure 2: Input choice validation

2. New Registration

2.1. Perfect Case / Successful Execution

Input student ID: se204826

This id is already exist

Input student ID: ce222222

Input student name (2 -> 20 characters): nguyen bao ngoc

Input student phone number: 0976153455

Input student email (Domain name: @fpt): ngocbn@fpt.edu.vn

Input peak code (Format: MT[NO.Mountain] eg:MT01,MT13,..): mt01

Add student successfully!

Press enter to continue!

[Screenshot: Input test case - display notification of adding student successfully]

Student id	Name	Email	Phone	PeakCode	Fee
SE999999	Tran Quang Toan	toanquangtran@fpt.edu.vn	0351912773	MT01	3.900.000
CE231467	Quan	quan@fpt.edu.vn	0908263519	MT05	6.000.000
SE204826	Tran Mai Anh Khoa	tranmaianhkhoea@fpt.edu.vn	0351923456	MT01	3.900.000
QE123646	Erwin Smith	erwinsmith@fpt.edu.vn	0901283857	MT09	6.000.000

[Screenshot: Evidence]

Figure 3: Execution Results - Perfect Case for New Registration

2.2. Catch Exception Case / Input Validation

```

Input student ID: iaksjdkwo
Please enter again!
Input student ID: sssssaa
Please enter again!

Please enter again!
Input student ID:
-----
| Choose yes or no                                     |
-----
| 0.Yes                                                |
| 1.No                                                |
|                                                    |
-----

Note: The system will ask for whether continuing to input information after 3 times failed
Do you want of continue?: 0
Input student ID: se204826
Input student name (2 -> 20 characters): a
Please enter again!
Input student name (2 -> 20 characters): @
Please enter again!
Input student name (2 -> 20 characters): tran anh khoa
Input student phone number: 1029383
Please enter again!
Input student phone number: -1
Please enter again!
Input student phone number: =3
Please enter again!

-----
| Choose yes or no                                     |
-----
| 0.Yes                                                |
| 1.No                                                |
|                                                    |
-----

Note: The system will ask for whether continuing to input information after 3 times failed
Do you want of continue?: 0
Input student phone number: 0833312345
Input student email (Domain name: @fpt): aassw
Please enter again!
Input student email (Domain name: @fpt): ki@gmail.com
Please enter again!
Input student email (Domain name: @fpt): khoa@fpt.edu.vn
Input peak code (Format: MT[NO.Mountain] eg:MT01,MT13,..): mt1
Please enter again, Peak Code is not exists
Input peak code (Format: MT[NO.Mountain] eg:MT01,MT13,..): mt17
Please enter again, Peak Code is not exists
Input peak code (Format: MT[NO.Mountain] eg:MT01,MT13,..): mt01
Add student successfully!
Press enter to continue!

```

[Screenshot: Input validation failure, error handling alerts, and the Yes/No exit prompt]

Figure 4: Execution Results - Exception Handling for New Registration

3. Update Registration Information

3.1. Perfect Case / Successful Execution

3.1.1 Searching student

```
-----
| Main menu                                     |
|-----|
| 0.Exit the Program                          |
| 1.New Registration                          |
| 2.Update Registration Information            |
| 3.Display Registered List                   |
| 4.Delete Registration Information            |
| 5.Search Participants by Name               |
| 6.Filter Data by Campus                     |
| 7.Statistics of Registration Numbers by Location |
| 8.View List of mountain info                |
| 9.Save Data to File                         |
|-----|
```

Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: 2

[Screenshot: Before updating info]

Input id to update: se204826

[Screenshot: Input id to search and update]

```
-----
Student ID: SE204826
Name       : Tran Mai Anh Khoa
Phone      : 0833346152
Mountain   : MT01
Email      : tranmaianhkhoa@fpt.edu.vn
Fee        : 3.900.000
-----
```

```
-----
| Update student menu                         |
|-----|
| 0.Return to menu                           |
| 1.Update student name                       |
| 2.Update peak code for student              |
| 3.Update phone number                       |
| 4.Update email                             |
| 5.Change student id to update               |
|-----|
```

Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice:

[Screenshot: Found student - Update menu display]

3.1.2 Update name

```
-----  
Student ID: SE204826  
Name      : Tran Mai Anh Khoa  
Phone     : 0833346152  
Mountain  : MT01  
Email     : tranmaianhkhhoa@fpt.edu.vn  
Fee       : 3.900.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email      |  
| 5.Change student id to update |  
|                     |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: 1
Input student name: tran anh khang|

[Screenshot: Input new name]

```
-----  
Student ID: SE204826  
Name      : Tran Anh Khang  
Phone     : 0833346152  
Mountain  : MT01  
Email     : tranmaianhkhhoa@fpt.edu.vn  
Fee       : 3.900.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email      |  
| 5.Change student id to update |  
|                     |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: |

[Screenshot: Evidence of saving new name]

3.1.3 Update phone number

```
-----  
Student ID: SE204826  
Name      : Tran Anh Khang  
Phone     : 0833346152  
Mountain  : MT01  
Email     : tranmaianhkhhoa@fpt.edu.vn  
Fee       : 3.900.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email      |  
| 5.Change student id to update |  
|                     |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: 3
Input phone number: 0901726354|

[Screenshot: Input new phone number]

```
-----  
Student ID: SE204826  
Name      : Tran Anh Khang  
Phone     : 0901726354  
Mountain  : MT02  
Email     : tranmaianhkhhoa@fpt.edu.vn  
Fee       : 6.000.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email      |  
| 5.Change student id to update |  
|                     |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: |

[Screenshot: Evidence of saving phone number - Tuition fee changed due to phone number]

3.1.4 Update peak code

```
-----  
Student ID: SE204826  
Name      : Tran Anh Khang  
Phone     : 0901726354  
Mountain  : MT01  
Email     : tranmaianhkhhoa@fpt.edu.vn  
Fee       : 6.000.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email      |  
| 5.Change student id to update |  
|                     |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed

Input your choice: 2

Input peak code (Format: MT[NO.Mountain] eg:MT01,MT13,..): mt02|

[Screenshot: Input new peak code]

```
-----  
Student ID: SE204826  
Name      : Tran Anh Khang  
Phone     : 0901726354  
Mountain  : MT02  
Email     : tranmaianhkhhoa@fpt.edu.vn  
Fee       : 6.000.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email      |  
| 5.Change student id to update |  
|                     |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed

Input your choice: |

[Screenshot: Evidence of saving peak code]

3.1.5 Update email

```
-----  
Student ID: SE204826  
Name      : Tran Anh Khang  
Phone     : 0901726354  
Mountain  : MT02  
Email     : tranmaianhkhhoa@fpt.edu.vn  
Fee       : 6.000.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email     |  
| 5.Change student id to update |  
|                   |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: 4
Input email: trananhkhang@fpt.edu.vn

[Screenshot: Input new email]

```
-----  
Student ID: SE204826  
Name      : Tran Anh Khang  
Phone     : 0901726354  
Mountain  : MT02  
Email     : trananhkhang@fpt.edu.vn  
Fee       : 6.000.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email     |  
| 5.Change student id to update |  
|                   |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: |

[Screenshot: Evidence of saving new email]

Figure 5: Execution Results - Successful Registration Information Update

3.2. Catch Exception Case / Data Validation

3.2.1 Searching student

```
Input id to update: nas
Please enter again!
Input id to update: sss
Please enter again!
Input id to update: se00
Please enter again!
```

```
-----
| Choose yes or no                                     |
| 0.Yes                                                |
| 1.No                                                |
|-----|
```

Note: The system will ask for whether continuing to input information after 3 times failed
Do you want of continue?: 0
Input id to update: se204826|

[Screenshot: Input wrong id format - display Yes/No exit after 3 attempts]

```
-----
Student ID: SE204826
Name      : Tran Anh Khang
Phone     : 0901726354
Mountain  : MT02
Email     : trananhkhang@fpt.edu.vn
Fee       : 6.000.000
-----
```

```
-----
| Update student menu                                     |
| 0.Return to menu                                       |
| 1.Update student name                                 |
| 2.Update peak code for student                       |
| 3.Update phone number                                 |
| 4.Update email                                         |
| 5.Change student id to update                         |
|-----|
```

Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: a
Please enter again!
Input your choice: s
Please enter again!
Input your choice: |

[Screenshot: Found student - Update menu display - Catch exception/wrong choice]

3.2.2 Update name

```
-----  
Student ID: SE204826  
Name      : as  
Phone     : 0901726354  
Mountain  : MT02  
Email     : trananhkhang@fpt.edu.vn  
Fee       : 6.000.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email      |  
| 5.Change student id to update |  
|                     |  
-----
```

```
Note: The system will ask for whether continuing to input information after 3 times failed  
Input your choice: 1  
Input student name: 1234  
Please enter again!  
Input student name: -123&*()  
Please enter again!  
Input student name: tran mai anh khoa
```

[Screenshot: Input new name - Catch wrong name format]

3.2.3 Update phone number

```
-----  
Student ID: SE204826  
Name      : tran mai anh khoa  
Phone     : 0901726354  
Mountain  : MT01  
Email     : tranmaianhkhhoa2006@fpt.edu.vn  
Fee       : 6.000.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email      |  
| 5.Change student id to update |  
|                     |  
-----
```

```
Note: The system will ask for whether continuing to input information after 3 times failed  
Input your choice: 3  
Input phone number: 9999999999182  
Please enter again!  
Input phone number: ddddasss  
Please enter again!  
Input phone number: 0833312346
```

[Screenshot: Input wrong phone number format]

3.2.4 Update peak code

```
-----  
Student ID: SE204826  
Name      : tran mai anh khoa  
Phone     : 0901726354  
Mountain  : MT02  
Email     : trananhkhang@fpt.edu.vn  
Fee       : 6.000.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email      |  
| 5.Change student id to update |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed

Input your choice: 2

Input peak code (Format: MT[NO.Mountain] eg:MT01,MT13,..): mt1

Please enter again, Peak Code is not exists

Input peak code (Format: MT[NO.Mountain] eg:MT01,MT13,..): mt19

Please enter again, Peak Code is not exists

Input peak code (Format: MT[NO.Mountain] eg:MT01,MT13,..): mt01

[Screenshot: Input wrong peak code]

3.2.5 Update email

```
-----  
Student ID: SE204826  
Name      : tran mai anh khoa  
Phone     : 0901726354  
Mountain  : MT01  
Email     : trananhkhang@fpt.edu.vn  
Fee       : 6.000.000  
-----
```

```
-----  
| Update student menu |  
-----  
| 0.Return to menu   |  
| 1.Update student name |  
| 2.Update peak code for student |  
| 3.Update phone number |  
| 4.Update email      |  
| 5.Change student id to update |  
|                     |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed

Input your choice: 4

Input email: khoa@gmail.com

Please enter again!

Input email: kaasdsasad

Please enter again!

Input email: ek09821

Please enter again!

```
-----  
| Choose yes or no |  
-----  
| 0.Yes            |  
| 1.No             |  
|                 |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed

Do you want of continue?: 0

Input email: tranmaianhkhoa2006@fpt.edu.vn

[Screenshot: Input wrong mail format]

4. Display Registered List

Student id	Name	Email	Phone	PeakCode	Fee
SE999999	Tran Quang Toan	toanquangtran@fpt.edu.vn	0351912773	MT01	3.900.000
CE231467	Quan	quan@fpt.edu.vn	0908263519	MT05	6.000.000
SE204826	Tran Mai Anh Khoa	tranmaianhkhoa@fpt.edu.vn	0351923456	MT01	3.900.000
QE123646	Erwin Smith	erwinsmith@fpt.edu.vn	0901283857	MT09	6.000.000

[Screenshot: When there is at least 1 student]

No students have registered yet. Press enter to continue!

[Screenshot: System messaging showing an empty state handling notification]

5. Delete Registration Information

5.1. Perfect Case / Successful Execution

Input student id to delete: se204826

```
-----  
Student ID: SE204826  
Name      : Tran Mai Anh Khoa  
Phone     : 0351923456  
Mountain  : MT01  
Email     : tranmaianhkhoa@fpt.edu.vn  
Fee       : 3.900.000  
-----
```

Are you sure?

```
-----  
| Choose yes or no |  
-----
```

```
| 0.Yes |  
| 1.No  |  
|       |  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed

Your choice: 0

Deleted successfully!

[Screenshot: Profile located, user confirms deletion request, and system confirms removal]

No students have registered yet.Press enter to continue!

[Screenshot: Evidence of deletion successfully]

Figure 6: Execution Results - Successful Registration Removal

5.2. Catch Exception Case / Missing Record

Input student id to delete: se204826

```
-----  
Student ID: SE204826  
Name      : Tran Mai Anh Khoa  
Phone     : 0351923456  
Mountain  : MT01  
Email     : tranmaianhkhoa@fpt.edu.vn  
Fee       : 3.900.000  
-----
```

Are you sure?

```
-----  
| Choose yes or no |  
-----
```

```
| 0.Yes  
| 1.No  
|  
-----
```

Note: The system will ask for whether continuing to input information after 3 times failed
Your choice: ass
Please enter again!
Your choice: -dk&^%*
Please enter again!
Your choice: ???/
Please enter again!
Your choice: 1
You denied to delete!

[Screenshot: Validation error response stating that the specified ID does not exist]

Figure 7: Execution Results - Exception Handling for Missing Deletion Target

6. Search Participants by Name

6.1. Perfect Case / Successful Execution

Input name to search: tran

Student id	Name	Email	Phone	PeakCode	Fee
SE999999	Tran Quang Toan	toanquangtran@fpt.edu.vn	0351912773	MT01	3.900.000
SE204826	Tran Mai Anh Khoa	tranmaianhkhoa@fpt.edu.vn	0351923456	MT01	3.900.000

Figure 8: Successfully found student by inputting partial name

6.2. Catch Exception Case / Unmatched Search

Input name to search: tran

There is no student with the similar name

Figure 9: None of student matches partial name

7. Filter Data by Campus

7.1. Perfect Case / Successful Execution

Input campus code [SE|CE|QE|HE|DE]: se

Student id	Name	Email	Phone	PeakCode	Fee
SE999999	Tran Quang Toan	toanquangtran@fpt.edu.vn	0351912773	MT01	3.900.000
SE204826	Tran Mai Anh Khoa	tranmaianhkhoa@fpt.edu.vn	0351923456	MT01	3.900.000

Figure 10: Successfully found

7.2. Catch Exception Case / Empty or Malformed Code

```

Input campus code [SE|CE|QE|HE|DE]: ak
Please enter again!
Input campus code [SE|CE|QE|HE|DE]: lo
Please enter again!
Input campus code [SE|CE|QE|HE|DE]: we
Please enter again!

```

```

-----
| Choose yes or no                                     |
-----
| 0.Yes                                                |
| 1.No                                                |
-----

```

```

Note: The system will ask for whether continuing to input information after 3 times failed
Do you want of continue?: 0
Input campus code [SE|CE|QE|HE|DE]: se

```

Student id	Name	Email	Phone	PeakCode	Fee
SE999999	Tran Quang Toan	toanquangtran@fpt.edu.vn	0351912773	MT01	3.900.000
SE204826	Tran Mai Anh Khoa	tranmaianhkhoea@fpt.edu.vn	0351923456	MT01	3.900.000

Figure 11: Try catch exception - Yes—No exit prompt

8. Statistics of Registration Numbers by Location

8.1. Perfect Case / Successful Execution

Student id	Name	Email	Phone	PeakCode	Fee
SE999999	Tran Quang Toan	toanquangtran@fpt.edu.vn	0351912773	MT01	3.900.000
CE231467	Quan	quan@fpt.edu.vn	0908263519	MT05	6.000.000
SE204826	Tran Mai Anh Khoa	tranmaianhkhoea@fpt.edu.vn	0351923456	MT01	3.900.000
QE123646	Erwin Smith	erwinsmith@fpt.edu.vn	0901283857	MT09	6.000.000

Figure 12: Current data

Peak Code	Number of student	Total cost
MT01	2	7.800.000
MT05	1	6.000.000
MT09	1	6.000.000

```

Note: The remaining mountain(s) that are/is not display, has its number of students is Zero

```

Figure 13: Statistic - Display data

8.2. Catch Exception Case / Guarded Statistics

There is no registion yet!

Figure 14: When there is no data or no registration

9. View List of Mountain Info

9.1. Perfect Case / Successful Execution

```

-----
Peak Code : MT01
Name      : Ham Rong Mountain
Province  : Lao Cai
Description : Located near the center of Sa Pa town. The highest point is 1850m above sea level.
-----
Peak Code : MT02
Name      : Doi Bo Mountain
Province  : Lao Cai
Description :
-----
Peak Code : MT03
Name      : Pha Luong Mountain
Province  : Son La
Description : Nearly 2000m above sea level. Pha Luong is known by many as the Roof of Moc Chau.
-----
Peak Code : MT04
Name      : Hon Vuon Mountain
Province  : Hue
Description :
-----
Peak Code : MT05
Name      : Da Do Mountain
Province  : Ninh Thuan
Description :
-----
Peak Code : MT06
Name      : Da Bia Mountain
Province  : Phu Yen
Description :
-----
Peak Code : MT07
Name      : Chu Hreng Mountain
Province  : Kon Tum
Description :
-----
Peak Code : MT08
Name      : Lang Biang Mountain
Province  : Lam Dong
Description :
-----
Peak Code : MT09
Name      : Ta Nang Mountain
Province  : Lam Dong
Description :
-----
Peak Code : MT10
Name      : Cam Mountain
Province  : An Giang
Description :
-----
Peak Code : MT11
Name      : Thi Vai Mountain
Province  : Vung Tau
Description :
-----
Peak Code : MT12
Name      : Dinh Mountain
Province  : Vung Tau
Description :
-----
Peak Code : MT13
Name      : Co Tien Mountain
Province  : Khanh Hoa
Description : The mountain has 3 peaks; when participating you only need to conquer peak 1; it is recommended to conquer peak 2 or 3 (with steeper slopes and more challenging cliffs)
-----

```

Figure 15: Display all mountain in file csv

10. Save Data to File

10.1. Execution Evidence & Screenshots

Save successfully!

Figure 16: Saving successfully notification

10.2. Binary File Verification

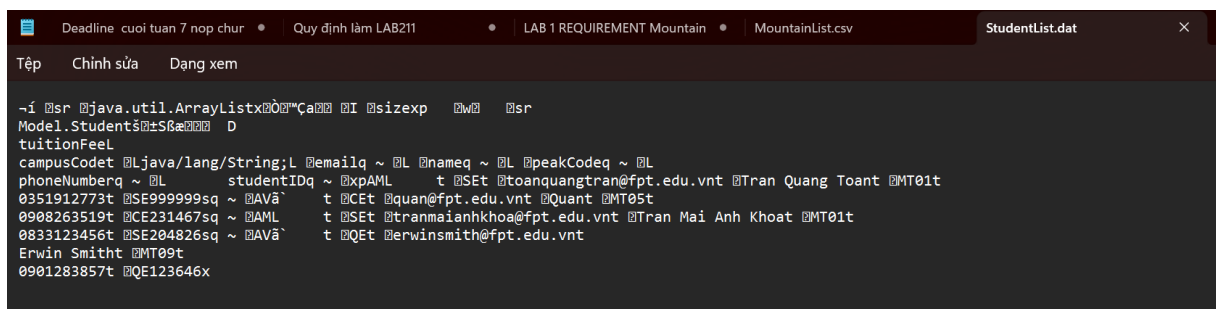


Figure 17: Evidence of saving file - Binary data - Serializable

11. UX/UI console

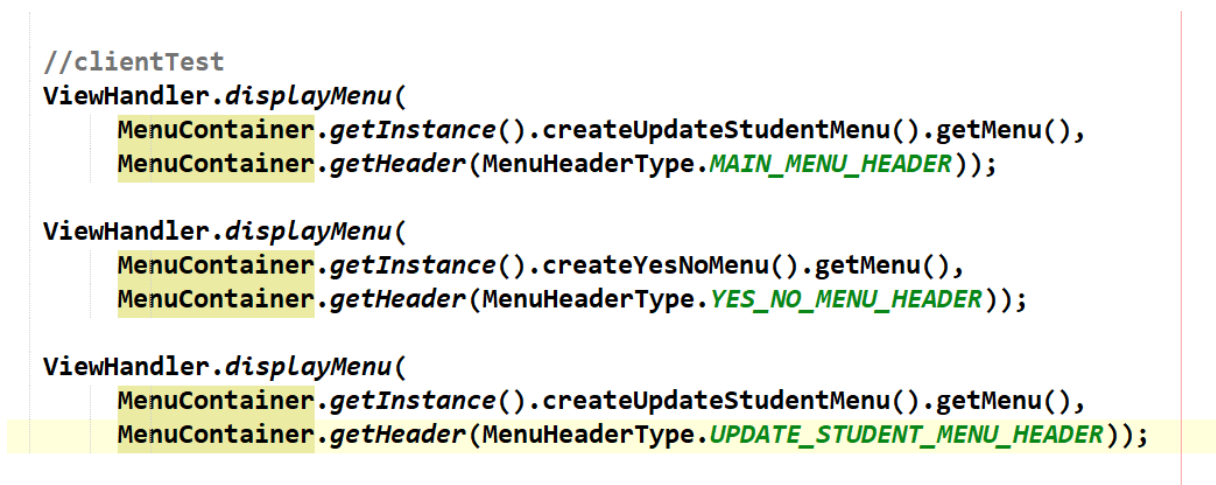


Figure 18: Using singleton pattern for menu, avoiding trash menu - Get Menu

```
-----
| Main menu                                     |
|-----|
| 0.Return to menu                             |
| 1.Update student name                       |
| 2.Update peak code for student              |
| 3.Update phone number                       |
| 4.Update email                             |
| 5.Change student id to update                |
|-----|
Note: The system will ask for whether continuing to input information after 3 times failed

-----
| Choose yes or no                             |
|-----|
| 0.Yes                                        |
| 1.No                                        |
|-----|
Note: The system will ask for whether continuing to input information after 3 times failed

-----
| Update student menu                           |
|-----|
| 0.Return to menu                             |
| 1.Update student name                       |
| 2.Update peak code for student              |
| 3.Update phone number                       |
| 4.Update email                             |
| 5.Change student id to update                |
|-----|
Note: The system will ask for whether continuing to input information after 3 times failed
BUILD SUCCESSFUL (total time: 0 seconds)
```

Figure 19: Menu demo

Do you want to save before leaving?

```
-----
| Choose yes or no                             |
|-----|
| 0.Yes                                        |
| 1.No                                        |
|-----|
Note: The system will ask for whether continuing to input information after 3 times failed
Input your choice: 0
Save successfully!
Press enter to continue!
```

Figure 20: Ask for saving data before leaving when did not save